

Cover letter — ATC 2026, Special Session *AI-Enabled 5G/6G Communication Systems*

Dear Editors of the 2026 International Conference on Advanced Technologies for Communications (ATC 2026),

We are pleased to submit our manuscript “Lightweight Stacked Ensemble for AI-Enabled 5G Network Intrusion Detection: A Latency–Accuracy Pareto Analysis” for consideration at ATC 2026, with a preferred placement in the Special Session *AI-Enabled 5G/6G Communication Systems*.

Topical fit. The manuscript addresses a problem that sits squarely inside the special session’s scope: deploying AI-driven network intrusion detection on the 5G access and edge plane under the latency constraints that govern in-line classification. We use the public 5G-NIDD benchmark—a 1.2-million-flow capture from a real 5G test network covering eight attack families across two base stations—to characterise both the classification quality and the latency–accuracy trade-off of a heterogeneous shallow stacked ensemble.

Contributions. The manuscript reports three contributions. First, we propose a heterogeneous shallow stacked ensemble that combines LightGBM, XGBoost, and a two-layer MLP through leakage-free out-of-fold probability stacking with a logistic-regression meta-learner. Second, we evaluate the ensemble under both a stratified random split and a cross-station split (BS1→BS2) that rebalances the per-class prior to isolate true generalisation across two physically distinct radio cells. Third, we trace the latency–accuracy Pareto frontier on a commodity 60-core CPU host across six batch sizes {1, 16, 64, 256, 1024, 4096} and publish a three-tier deployment recommendation table mapping operating points (latency budget, accuracy floor) to the configuration that meets them.

Originality and scope. The submission is original, has not been published elsewhere, and is not under consideration by any other venue. All four authors have read and approved the manuscript. The artefact (code, configuration, ten project seeds, and aggregation pipeline) is reviewer-runnable from a single command and is released under MIT, with a Zenodo DOI to be minted upon acceptance.

Suggested topics (ATC tracks). The work intersects the *AI-Enabled 5G/6G Communication Systems* special session and the *Network Operations and Management* main track. We respectfully request consideration for the special session as the primary fit.

We thank you and the reviewers for your time. We welcome any feedback that would strengthen the artefact or the manuscript for the proceedings.

Sincerely,
On behalf of all authors,
Van-Quynh Trinh
Corresponding author
Posts and Telecommunications Institute of Technology (PTIT)
quynhtr@ptithcm.edu.vn